

Principles, decisions & everything we built

The full record of the July 2026 design sprint: three clickable prototypes for Zaki's GitHub issues, the seven principles that shaped them, every design decision with its reason, the audit that hardened them, and the research spine underneath.

BUILT & DELIVERED

- 01 Prompts prototype** — write · try · batch · compare · compress · govern (Issue #1)
- 02 Inbox prototype** — feed + task queue, act-to-clear enforced in code (Issue #3)
- 03 Platform prototype** — control plane + audited Enter-org (Issues #4/#2)
- 04 Execution plan** · domain guide + jargon · 3-round audit log · GitHub drafts · reusable skill

Project Esberi / Keystone

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Branch prompt-management-v1

◆ keystone

Keystone Design Report – what we built, the principles, and the decisions

2–3 July 2026 · Amaan (design) · covering the prompt-management engagement and Zaki's four GitHub issues. This is the narrative record: every deliverable, every principle we designed by, every decision with its reason, and the method used to get there.

1. Executive summary

In two days the engagement went from a kickoff huddle to **three client-ready, clickable prototypes** plus the research and documentation spine behind them:

DELIVERABLE	WHAT IT IS	WHERE
Execution plan	Master plan: phases, screen inventory, PRD-vs-repo gap audit, seeded GitHub questions	Keystone-PromptManagement/00-EXECUTION-PLAN.md
Domain guide + jargon dictionary	"Become a domain expert" doc: prompt management from zero, market state (Langfuse/Phoenix verified against live docs), 14 future assumptions → design decisions, ~120-term glossary	Keystone-PromptManagement/03-DOMAIN-GUIDE.md (+ PDF)
Prompts prototype (Issue #1)	The full write → try → compare → compress → govern loop, Maya scenario seeded	Keystone-Prototypes/Keystone-Prompts-Prototype.html
Inbox prototype (Issue #3)	Cross-project feed + task queue with the act-to-clear mechanic enforced in code	Keystone-Prototypes/Keystone-Inbox-Prototype.html
Platform prototype (Issues #4 + #2 tie-in)	System Admin control plane with the audited Enter-org assume-role journey	Keystone-Prototypes/Keystone-Platform-Prototype.html
Briefs + style contract	Zaki's briefs verbatim + the binding visual/interaction contract all three prototypes share	Keystone-Prototypes/briefs/
Audit log	Three formal ui-ux-audit rounds: findings, severities, fixes, deliberate non-changes	Keystone-Prototypes/AUDIT-NOTES.md
GitHub comment drafts	Ready-to-paste replies for issues #1–#4, including the open backend questions	Keystone-Prototypes/GITHUB-COMMENTS.md
Reusable skill	The whole workflow packaged: research → build → audit, one command for future briefs	~/.claude/skills/brief-to-prototype/

Every prototype is a single self-contained HTML file — no framework, no build step, no backend, no network calls — exactly what Zaki asked for ("mock prototype, no real backend wiring"). All three passed syntax validation, ran with zero console errors, and were exercised end-to-end in the browser in dark **and** light themes.

2. The journey

WHEN	WHAT HAPPENED
Jul 2, morning	Huddle with Zaki: scope (prompt management first), method (3–5 pressure-tested principles), differentiators (LLMLingua compression + model flexibility), process (GitHub not Slack, daily syncs, mock-only)
Jul 2	Execution plan written; <code>keystone-main</code> audited end-to-end — the PRD-vs-repo gap table became the backbone for both the mock scope and the GitHub questions

WHEN	WHAT HAPPENED
Jul 2	Domain guide written; Langfuse + Arize Phoenix facts verified against their live docs (not memory); jargon dictionary added on request
Jul 2, evening	Zaki filed four GitHub issues (#1 Prompt Management, #2 Org Admin — no brief text, #3 Inbox, #4 System Admin), all assigned; every brief specifies "shadcn/ui + lucide, default styling, dark developer console" — overriding the earlier Terminal-TTY direction
Jul 3	Briefs persisted verbatim + style contract authored; Lazyweb research pass; parallel build agents launched — all four killed by the org's monthly spend limit — so all three prototypes were built inline instead
Jul 3	Pencil file checked (org/admin screens exist for #2; Inbox = zero coverage; new-spec System Admin frames missing — deferred to a Pencil session)
Jul 3	Browser verification (interactions exercised, 2 layout fixes) → formal 3-round ui-ux-audit (4 findings fixed, 2× P0) → GitHub comment drafts → this report

3. Ground truth we established first

Design decisions were made against evidence, not assumptions:

- **The PRD is target-state; the repo is minimal.** `keystone-main` stores a prompt as `{name, description, template, version}` — integer versions with non-destructive restore, f-string `{variable}` (not mustache), `{{secret:KEY}}` tokens, flow-node pinning. It has **no** version-tags, labels, prompt references, compression, direct prompt runs, dollar costs, or Connections. Everything the prototypes mock beyond that is deliberate target-state design, and each gap is a drafted GitHub question.
- **The PRD is a remix of Langfuse and Phoenix** (verified live): Langfuse's text/chat kinds, `{{variables}}`, labels-as-deployment-pointers, protected labels, `@@@prompt@@@` composability; Phoenix's fixed tag set (production/staging/development) and playground energy. Keystone's two bets that neither has: **LLMLingua compression with visible token economics** and **builder-owned credentials with first-class local models**.
- **Reference patterns from research:** Linear/Front-style split list + reading pane with inline row actions (Inbox); status-badged metadata tables (platform consoles).

4. The seven principles we designed by

The huddle rule: principles must be able to lose an argument — each one lists where it forced a real decision in the build.

- P1 • The loop is sacred.** Write → try → inspect → fix → ship, with the common path one click. *Visible:* quick-run lives inside Edit (no mode switch to try); hover-Run on every list row; ⌘K reaches any prompt/action in two keystrokes; Compare hands winners straight to Promote.
- P2 • Governed by default, not by chore.** Versioning and audit are byproducts of normal work, never a task. *Visible:* every save auto-mints a version ("Saved as v13"); restore is non-destructive ("Restored v9 as v13"); promotion moves a pointer via an explicit, confirmed dialog; every Platform action appends a live audit row.
- P3 • One way to do a thing.** A Tabs trigger switches mode · a toolbar icon acts on the whole object · an ... menu acts on the one item clicked · every pop-up is a centered dialog. *Visible:* the same anatomy across all three prototypes — a reviewer flipping between them sees one product. This is Zaki's "one rule," implemented literally.
- P4 • Show, don't hide, what happened.** Every result is inspectable to its raw internals. *Visible:* Inspect opens the fully-assembled prompt, raw request/response JSON, token breakdown (in/out/cached), cost, latency, and a step timeline; the Platform audit shows before→after on every event.
- P5 • Boundaries are features.** The builder owns keys; the operator can't read tenants; a decision can't be dismissed as a notification. *Visible:* secrets never render anywhere (operator included); Enter-org demands re-auth and wears a persistent audited banner; a pending actionable has no archive button — and clicking where it would be explains why.

P6 · Cost is a first-class signal (*prompt-specific*). *Visible*: the Token-budget card computes size/cost live while typing; every run row carries tokens · cost · latency; Compress leads with Before/After/Saved/ratio/cost-saved — the differentiator gets the most honest arithmetic on the surface.

P7 · Earn every component (*craft*). Nothing ships because a library has it. *Visible*: no stats bar on the list (brief said not yet); no side drawers anywhere; destructive friction scales with risk (archive = one confirm; delete = re-auth + type-the-name with the button gated on an exact match); the Compress "Apply" stays disabled until there is genuinely something to apply.

5. The craft floor (applied via the ui-ux-audit framework)

Beneath the product principles, every screen was held to the measurable floor from Practical UI + Refactoring UI: text ≥4.5:1 and UI borders ≥3:1 contrast (both themes); visible keyboard focus everywhere; one primary button per view; destructive actions calm until the decision point; never colour-only meaning (every dot pairs with text); designed empty states on every list; prefers-reduced-motion honored; numbers right-aligned tabular; hierarchy carried by weight and muted-foreground rather than ever-bigger text. Four violations were found in the audit (two P0) and fixed — the full table with measurements is in AUDIT-NOTES.md .

6. Design decisions and why

Global (all three prototypes)

DECISION	WHY
shadcn New York, default styling, dark developer console — not the Terminal-TTY skin	Every one of Zaki's four briefs specifies it; the client's written instruction supersedes our internal iteration. The Admin prototype became the foundation; the Terminal skin remains the flow-editor exploration
One self-contained HTML file per surface, mock-only	Zaki's explicit ask ("token-efficient agent rewiring", no backend); also makes review one double-click
Shared style contract file before any code	Three surfaces built (partly in parallel) must read as one product; the contract pins tokens, shell anatomy, and the "one rule"
Hash router + screen registry, plain DOM	The Admin prototype's proven pattern; every screen is URL-addressable for review
Inline lucide-style SVG icons, zero CDN	Self-contained constraint; icons stay crisp in both themes
Mock data tells the brief's scenario story	Maya's triage/tone journey (#1), Sam's triage-in-30-seconds (#3), Northwind's recovery (#4) — every screen demonstrates the narrative the client wrote, not lorem ipsum
Dark default with a first-class light theme	The briefs say dark developer console; the foundation demands light parity — both AA-verified
Toasts + ⌘K + Esc semantics everywhere	The rhythm of the product: acknowledge every action, reach everything by keyboard

Prompts (#1)

DECISION	WHY
Auto-version on save (blur → "Saved as v13")	P2 made real: governance as a byproduct; also matches the repo's snapshot-on-edit behavior
Restore non-destructive ("Restore v9 → creates v13")	Matches shipped keystone-main semantics; history is never rewritten
{{variable}} syntax as the brief writes it, with a GitHub question flagged	The repo ships f-string {var} — a real conflict for engineering to settle, not for the mock to hide
Kind (chat/text) fixed at creation; toolbar toggle explains and points to Duplicate	The brief fixes the editor shape per kind; a silent convert would lie about the data model

DECISION	WHY
Env badges: production green · staging amber · dev blue · draft grey	The brief specifies these exact colors (App. B)
References as chips with live resolution (green found / red missing, seeded with one broken)	States must be honest — the unhappy path is designed, not hidden
Quick-run panel slides inside Edit (not a dialog)	The brief's one exception to centered dialogs; keeps the loop unbroken
Compress: methods as plain-language cards, protected tokens visually locked, Apply mints a version	The differentiator must build trust: show exactly what's removed, prove slots survive, make it instantly reversible
Compare limited to 8 visible rows when 50 loaded, labelled	Honesty over spectacle: the label says what's shown rather than faking infinite rendering
Batch seeds one deliberate misclassification	Gives Inspect a reason to exist in the demo — debugging from evidence

Inbox (#3)

DECISION	WHY
<code>item_class</code> drives everything: read-to-clear vs act-to-clear, enforced in the code paths	The brief's load-bearing rule; mark-read literally cannot resolve an actionable
Badge counts pending actionables , not unread	The badge answers "is anything waiting on me?" — unread noise would poison trust
No Inbox nav item; entry = badge on the account block	Brief App. A; the Inbox follows the person, not the project
Needs-action sorts by urgency (severity, then due), All by recency	A queue is not a timeline — the brief calls this out explicitly
One approval auto-resolves after ~20s ("Resolved by Dana Ortiz")	Demonstrates first-decision-wins/no-zombie-tasks live, the property that makes the queue trustworthy
Expired item shows "expired · missed", out of the queue but in the feed	Expiry never masquerades as done
Row-level Reject/Decline calm (outline); full emphasis only in the detail pane	Audit finding: repeated solid red down a queue is noise; destructive stays calm until the decision point
Placeholder Build screen included	Proves the "triage and hop back" journey — the Inbox is a surface you leave

Platform (#4, with #2 tie-in)

DECISION	WHY
Login → MFA step first; re-auth again for delete and enter-org	"Even the operator re-proves identity" — JTBD-5 verbatim
Delete disabled until suspended, with the hint inline in the menu	The lifecycle rule enforced where the user would break it, not in documentation
Soft delete with visible recovery deadline + Restore	Blast radius stated, reversibility designed (30-day window)
Enter-org = two-step (re-auth → audited-session consent) → persistent amber banner + Leave	The crossing must feel like a crossing; the banner makes "I am inside a tenant" impossible to forget
Entered view is governance-only: members, projects, promote, transfer — nothing else	Control plane vs data plane; "no authoring" and mask-forever secrets stated on the surface itself
Audit appends live rows for every session action; operator rows highlighted with a one-click filter	The "prove isolation" view — the answer to "can your operator read our data?"
Northwind seeded flagged "no reachable admin"	Makes the recovery journey (enter → promote → leave) demonstrable end-to-end

DECISION	WHY
☞K "Simulate fresh deployment"	Shows the day-zero empty state without deleting demo data by hand

7. How it was designed – the method

The three-skill pipeline (now packaged as the reusable `brief-to-prototype` skill): 1. **Research** (design-new / Lazyweb): quick pattern scans per surface + live-doc verification of Langfuse/Phoenix — research fills gaps but never overrides a detailed client brief. 2. **Build** (frontend-design): briefs persisted verbatim to files; a binding style contract authored first; one self-contained HTML per surface with a "what must work" checklist derived line-by-line from the brief's appendices; scenario-seeded mock data. 3. **Audit** (ui-ux-audit): three rounds — functional verification with real clicks, then dimension-by-dimension against the measurable floor, then re-verification. Findings logged with severity and measurements; deliberate non-changes recorded with reasons.

Verification discipline: every file's script extracted and parsed with `node --check` before any browser time; then served over local HTTP (file:// is blocked) with `?v=N` cache-busters bumped on every edit; console watched for runtime errors; the load-bearing interactions clicked, not assumed — approve-clears-queue, badge decrement, the full enter-org journey, the compress compute-then-apply handoff.

The fallback that saved the day: all four parallel build agents were killed mid-run by the org's monthly spend limit. The deliverables didn't slip — the three prototypes were built inline in the main session instead, using the same briefs, contract, and checklists the agents had been given. That resilience is now written into the skill.

Three design-refinement passes followed the audit (full log in `AUDIT-NOTES.md`, Rounds 4–6), all staying strictly inside Zaki's shadcn / dark-console contract: - *Round 4 — feel:* a shared elevation + motion layer (two-part tinted shadows, purposeful transitions, dialog/menu entrances, button press states); **streaming run output** so the model appears to "type" its answer with a terminal cursor; **drag-to-reorder** message segments; **count-up** on the compress savings; and the Inbox's signature move — **deciding animates the item out of the queue** with a badge pulse. - *Round 5 — states & completeness:* **skeleton loading states** on every list (the one missing state from Keystone's 4-states discipline); **custom scrollbars** matching the console aesthetic; **undo in toasts** for reversible actions (Practical UI's "always allow undo"); a **Compare winner tally**; and cross-prototype consistency polish. All reduced-motion-gated; all re-validated in the browser. - *Round 6 — shadcn New York fidelity + Org Admin:* preceded by fresh shadcn New York research and a written **gap analysis** (`GAP-ANALYSIS.md` — "what Zaki's contract wants vs what I originally planned"; the pre-contract plan assumed the Terminal-TTY skin + maximal PRD model, both overturned) and a fidelity spec (`briefs/SHADCN-NEWYORK-SPEC.md`). Produced `-v2.html` for the three surfaces (the material change: badges pill→`rounded-md` per authentic New York, radius ladder + elevation confirmed, **JS/interactions untouched**) and built the previously-missing **Org Admin prototype** (`Keystone-OrgAdmin-Prototype.html`, Issue #2): eight org-scope surfaces — Overview · Members (roles + invite + role matrix) · Projects · Secrets (write-once, masked-forever) · Connections · Billing (budget gate + rate card) · API keys · Audit. Verified dark + light.

8. What the client sees (selected screens)

captured live from the prototypes at 1440x900 · dark theme unless noted

acme-support

BUILD

Flows

Prompts

EVALUATE

Experiments

Datasets

OPERATE

Schedules

OBSERVE

Runs

MANAGE

Members

Settings

Amaan K
Builder

Prompts5 prompts · acme-support

Sort

Archived

New

NAME	KIND	LABELS	VERSION	ENV	LAST RUN	RUNS
support-triage	chat	team:supportpii	v12	production	2m ago	412
tone-rewriter	text	team:supporttone	v8	staging	1h ago	96
safety-guidelines	text	compliance	v4	production	—	3
escalation-detector	chat	experimental	v2	dev	3d ago	17
onboarding-writer	text	—	v1	draft	never	0

01 · Prompts – the landing list: every brief column, env badges in the specified colors, hover-Run, grouped row menu

acme-support

BUILD

Flows

Prompts

EVALUATE

Experiments

Datasets

OPERATE

Schedules

OBSERVE

Runs

MANAGE

Members

Settings

Amaan K
Builder

Prompts · support-triageproductionv12 · production

Edit

Run

Manage

SYSTEMstanding orders

You are a support assistant for Acme Corp. Follow the shared safety rules at all times. safety-guidelines Be precise and never invent ticket details.

placeholder · chat_history – injects a whole conversation at run time

USERthe request

Classify the intent of this message: {{message}} – into exactly one of: billing, technical, account, or other. Reply with only the category name.

Add message

Add placeholder

OUTPUT

MessageTextJSON

What this prompt returns to the flow that calls it.

INPUTS

{{message}}

STRING

chat_history

MESSAGES

REFERENCES

safety-guidelinesFOUND

TOKEN BUDGET

claude-sonnet-4-6

65 tok0% of 200kctx

~\$0.0002 / run

02 · Edit mode – message segments with role chips, the safety-guidelines reference chip resolved green, live {{variable}} inputs and the amber-free token budget

acme-support

BUILD

Flows

Prompts

EVALUATE

Experiments

Datasets

OPERATE

Schedules

OBSERVE

Runs

MANAGE

Members

Settings

Amaan K
Builder

Prompts · tone-rewriter · staging · v8 · staging

T ⓘ ⚙ ⌂ ↺ ⌵ 🔍 Edit Run Manage

Compress prompt

shortens the prompt; meaning kept, slots protected

LLMLingua

Classic budget-driven compression. Balanced quality and shrink.

LongLLMLingua

Question-aware — keeps what matters for a specific question in long context.

LLMLingua-2

Fast task-agnostic distillation. Best default for prompts like this.

Keep-rate

Target tokens

keep ~60%

Preserve digits

Compress

BEFORE

105 tok

AFTER

63 tok

SAVED

-42 tok

SHRINK

40%

COST SAVED

\$0.02 / 1k runs

You are an expert customer-communication editor for the Acme support team. Rewrite the following message in a `{{tone}}` tone, keeping the meaning, the facts, and any ticket numbers exactly the same. Always make sure to preserve product names, dates, and monetary amounts precisely as they were written. Do not add new information, do not remove important details, and keep the reply reasonably concise so that customers are able to read it quickly and easily. `[[ref:safety-guidelines]]` Message to rewrite: `{{message}}`

Cancel

Apply compression

03 · Compress – the differentiator: three plain-language methods, protected slots highlighted, struck-through removals, honest before/after economics

acme-support

BUILD

Flows

Prompts

EVALUATE

Experiments

Datasets

OPERATE

Schedules

OBSERVE

Runs

MANAGE

Members

Settings

Amaan K
Builder

Prompts · support-triage · production · v12 · production

⌵ ⌵ 🔍 Edit Run Manage

Details

Name

support-triage

Description

Triage classifier for Acme help-desk tickets

Labels

team:support pii Manage tags

Versions

v12

production

Tightened category instruction; reply with category only

2m ago · Maya

...

v11

—

Original classification wording

2d ago · Maya

...

v10

staging

Added safety-guidelines reference

5d ago · Amaan

...

v9

—

Added chat_history placeholder

1w ago · Maya

...

v8

dev

System persona rewrite

2w ago · Maya

...

v7

—

Initial import from playground

3w ago · Amaan

...

Associated datasets

helpdesk-tickets-2026-06

50 ROWS · READ-ONLY

Access

Amaan K

amaan@esperi.com

Owner

...

Maya Chen

maya@acme.com

Editor

...

04 · Manage – versions with environment pointers, diff/restore/promote, datasets, access roles

acme-support

PINNED

support-triage

BUILD

Flows

Prompts

EVALUATE

Experiments

Datasets

OBSERVE

Runs

MANAGE

Members

Settings

Sam K

Builder · 3 projects

Inbox

Needs action 5

All

Unread

Archived

All types

All projects

Q

Mark all read

Approval needed — data-cleanup is paused before a production write

internal-tools · flow data-cleanup · paused run #2214 · 18m ago

pending

due in 4h

Approve

Reject

API token acme-ci expires in 2 days

acme-support · token acme-ci · credentials · 5h ago

pending

expires in 2d

Renew

Acknowledge

Approval needed — deploy-gate before publishing churn model

acme-support · flow deploy-gate · paused run #2199 · 2h ago

pending

due in 1d

Approve

Reject

Leo Park requested access to internal-tools

internal-tools · project access · Leo Park · 3h ago

pending

Grant

Deny

Maya Chen invited you to churn-radar

churn-radar · project · Maya Chen · 32m ago

pending

Accept

Decline

Maya Chen assigned you a comment on support-triage v12

acme-support · support-triage v12 · Maya Chen · 1h ago

pending

Resolve

Open

Select an item to read it here.

Decisions happen in place — deep-links for depth.

05 · Inbox – the Needs-action queue: urgency-sorted, accent-barred actionables with calm inline decisions, badge = pending count

KEYSTONE

PLATFORM

Organizations

Platform Settings

Platform Audit

Platform · Organizations

Search orgs...

+ Create org

Q

ORGANIZATIONS

8

ACTIVE

5

PENDING SETUP

1

SUSPENDED / DELETING

2

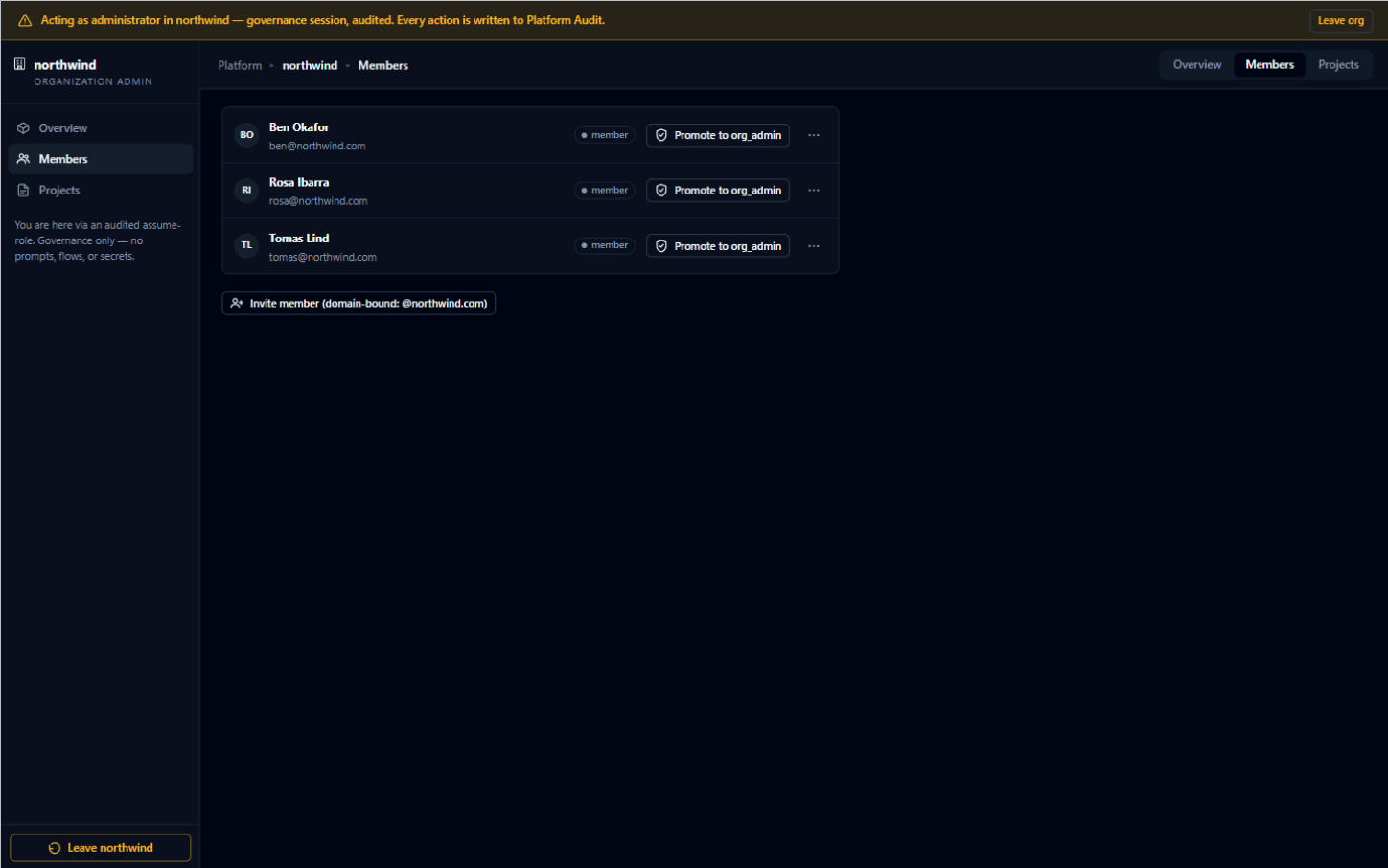
NAME	EMAIL DOMAIN	MEMBERS	PROJECTS	STATUS	CREATED	ACTIVITY
acme	acme.com	42	7	active	Jan 12, 2026	2m ago
northwind	northwind.com	3	3	active	Mar 3, 2026	3d ago
initech	initech.io	0	0	pending setup	Jul 1, 2026	—
globex	globex.co	17	5	suspended	Sep 20, 2025	3w ago
umbrella	umbrella.health	11	4	pending delete	Jun 2, 2025	6w ago
stark-labs	stark.dev	23	9	active	Nov 2, 2025	1h ago
wayne-corp	wayne.co	15	4	active	Feb 8, 2026	yesterday
hooli	hooli.xyz	51	12	active	Aug 15, 2025	4h ago

Metadata only at this altitude — an org's prompts, flows, and secrets are tenant data and are never listed here.

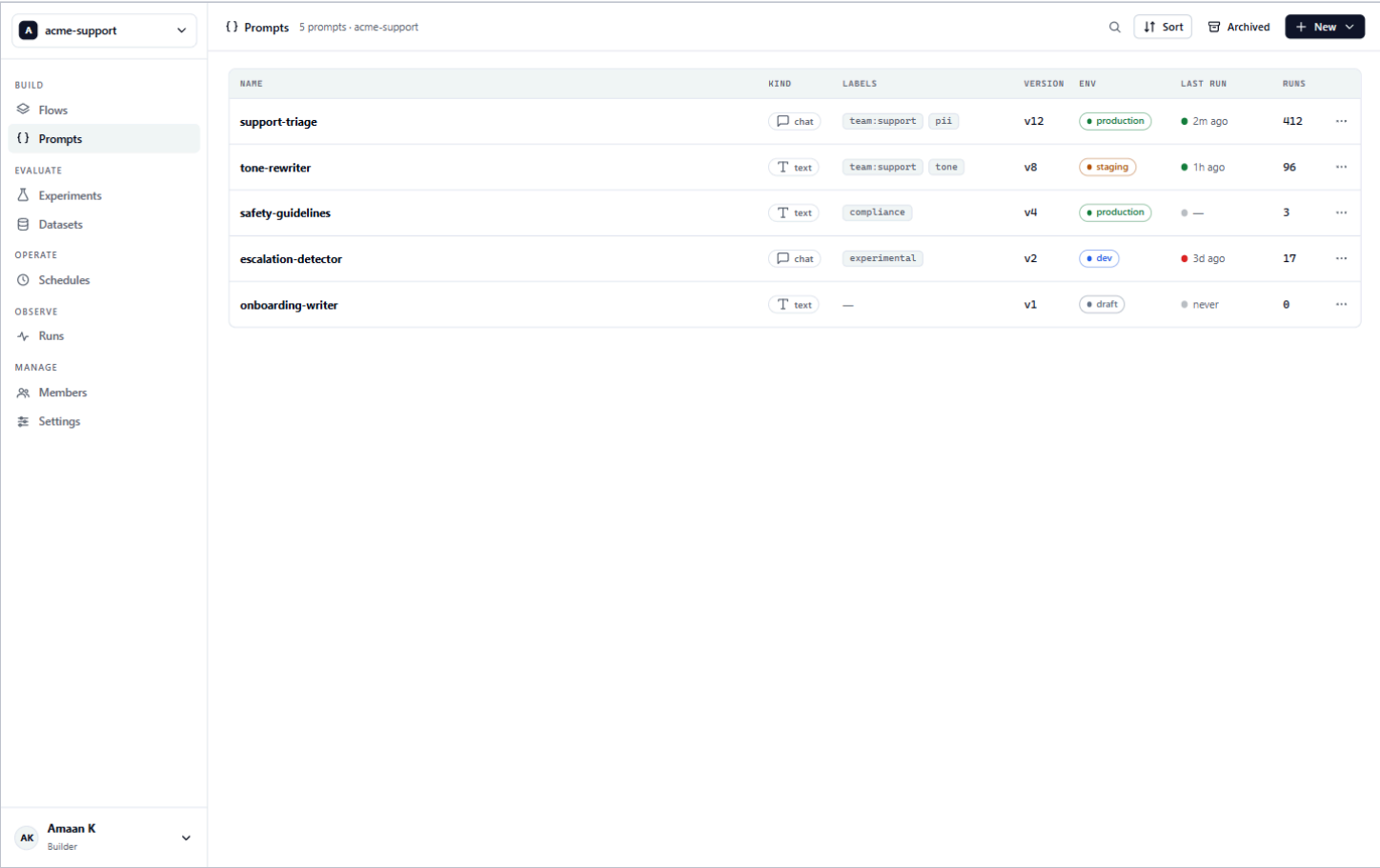
Operator

operator@esberl.com

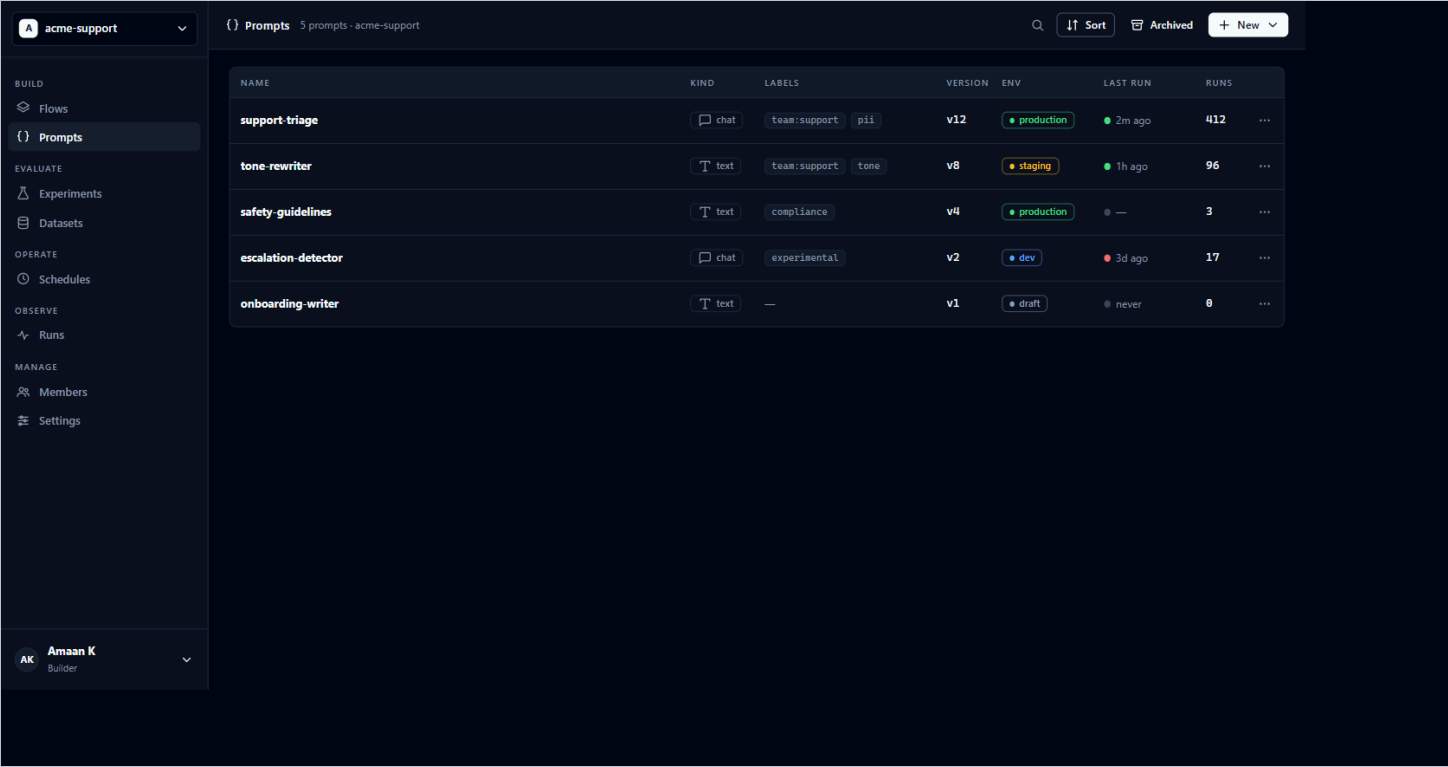
06 · Platform – the organizations portfolio: all four lifecycle states, the northwind recovery flag, metadata-only altitude



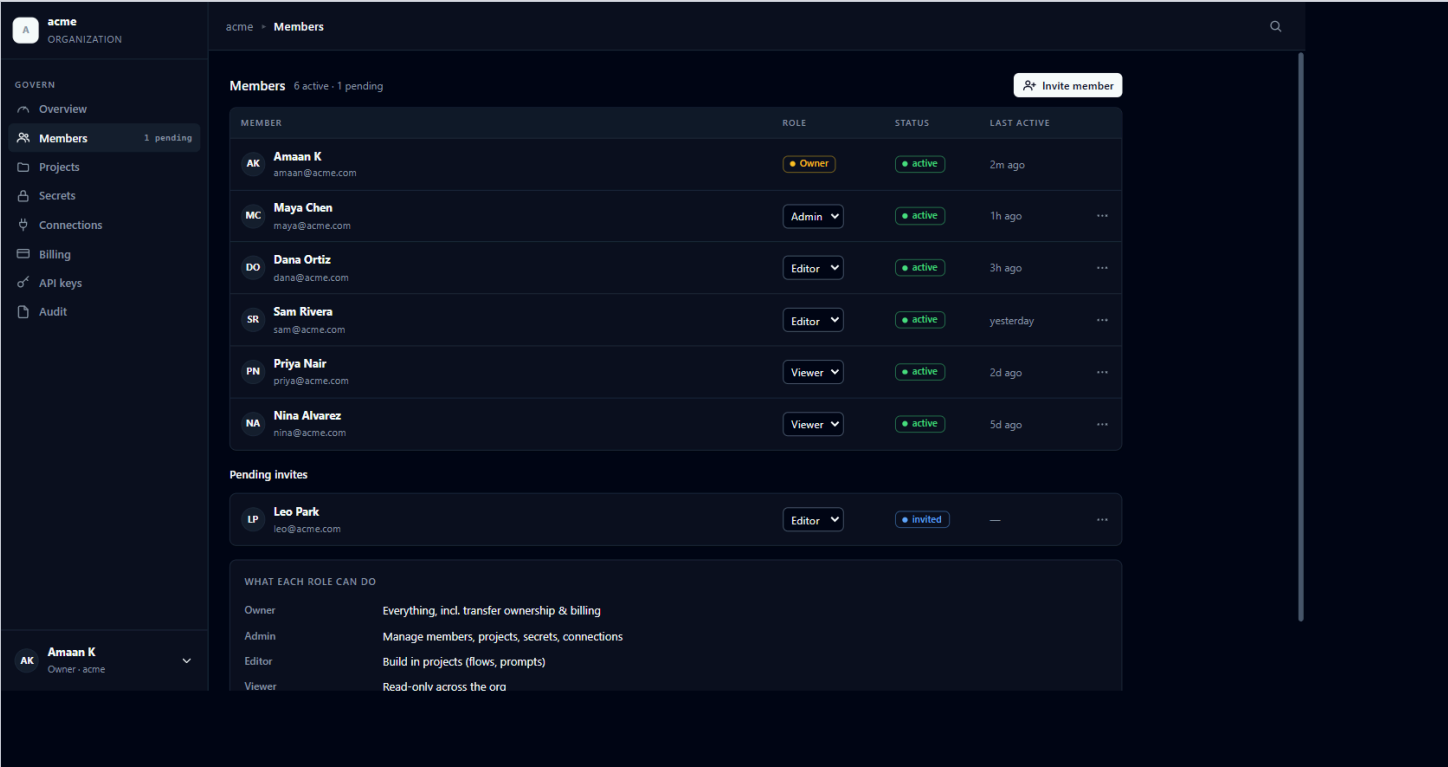
07 · Enter-org assume-role – persistent audited banner, governance-only sidebar, promote-to-org_admin recovery path



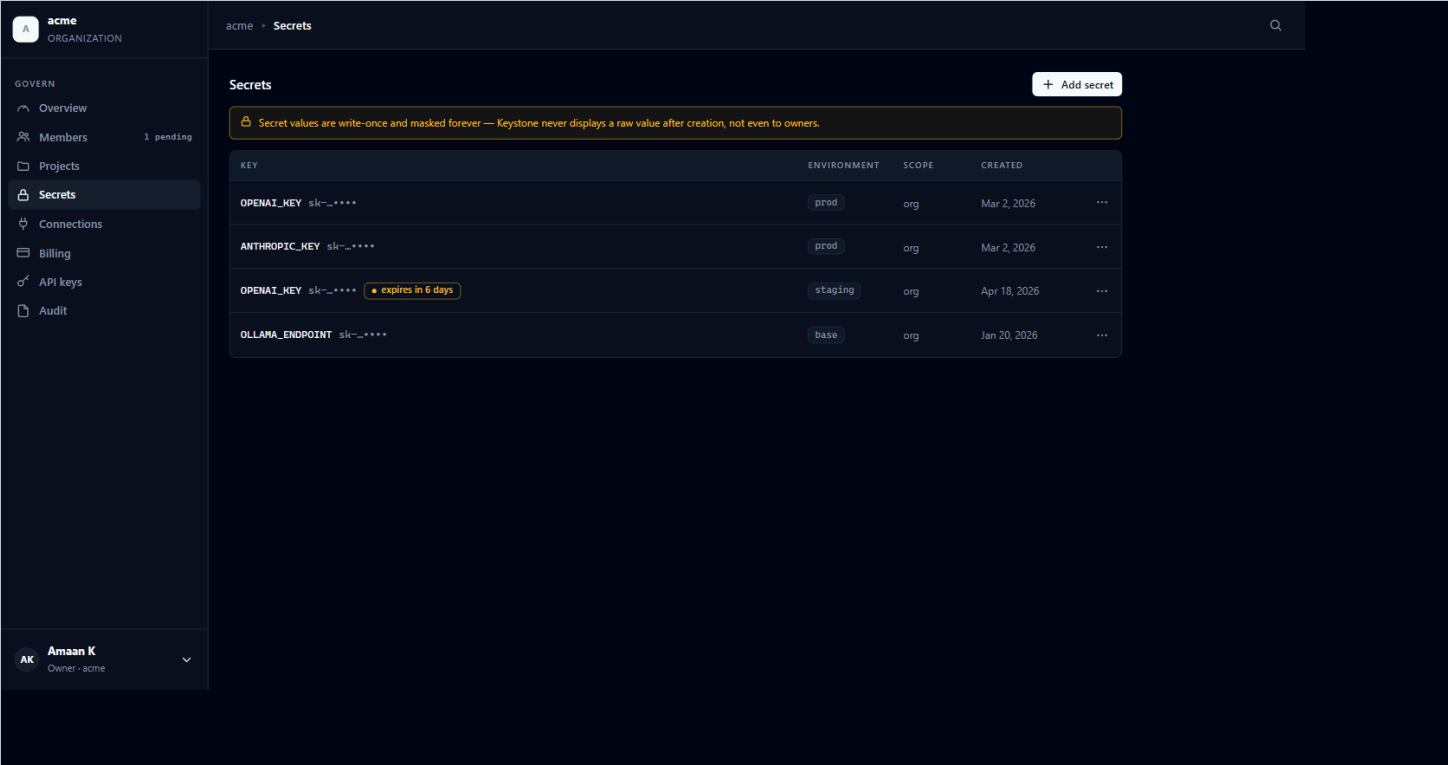
08 · Light theme parity – same hierarchy, AA contrast, one token system



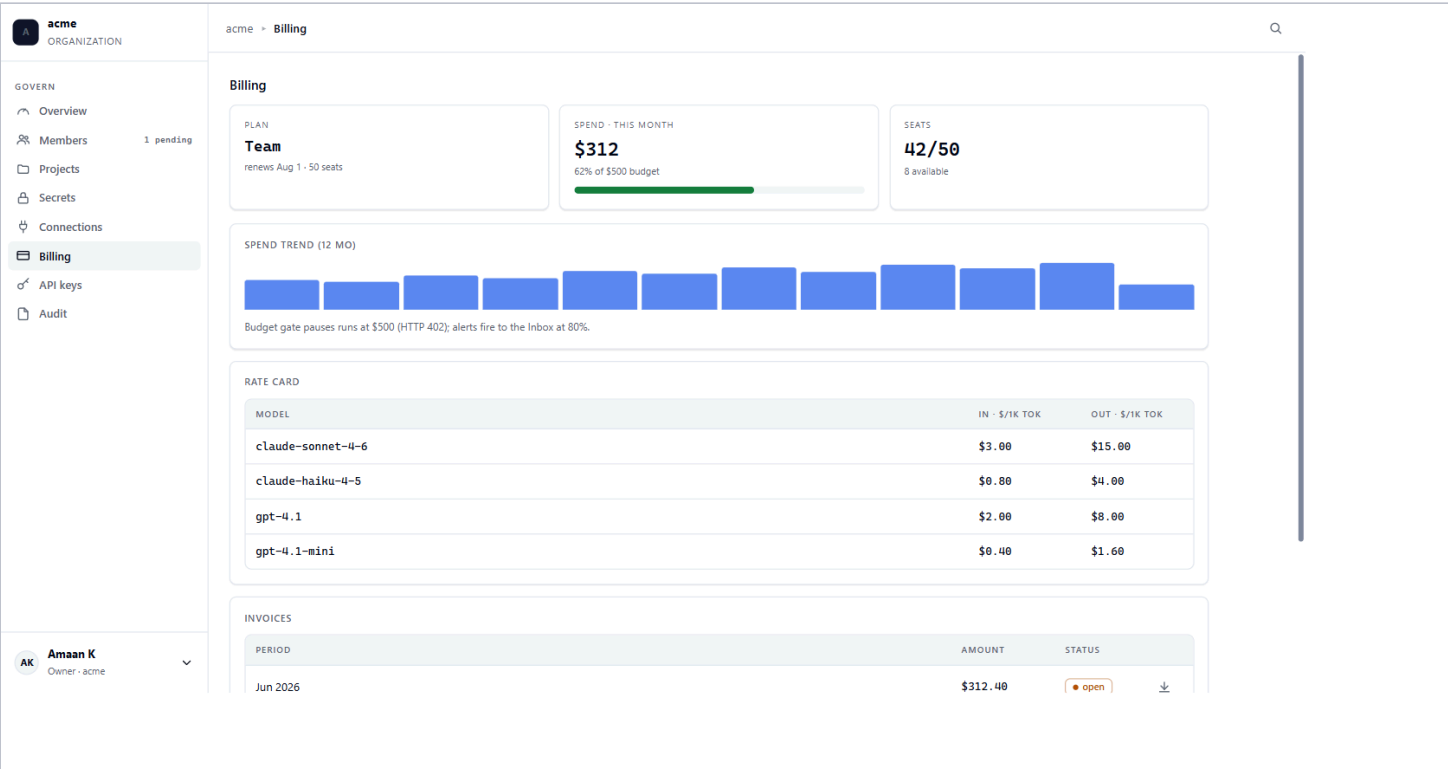
09 · Round 6 – shadcn New York fidelity: badges refined from pills to rounded-md rectangles (Prompts v2)



10 · New Org Admin (Issue #2) – Members: Owner/Admin/Editor/Viewer roles, invite, pending invites, role matrix



11 · Org Admin – Secrets: write-once, masked-forever values with an expiry warning (the governance centerpiece)



12 · Org Admin – Billing in light theme: budget gate, spend trend, rate card, invoices

9. What remains

1. **Pencil frames** — Inbox and new-spec System Admin screens in `Keystone-Admin-Design.pen` (gap-check done; org-admin coverage exists for #2; best done in a fresh session with Pencil open).
2. **Post the GitHub comments** (drafted, not sent) — ideally after committing the prototypes to the `prompt-management-v1` branch so the comments can link to them.

3. **Issue #2 brief** — waiting on Zaki's Organization Admin text; the Enter-org governance view already covers the structural core.
4. Daily-sync talking points: the three backend questions from issue #1 (variable syntax, chat storage shape, price table).